

1

Given a linear model with explicit bias term $g(\mathbf{x}; \mathbf{w}, b)$, we want to obtain an update rule for b similar to the obtained update rule for $\mathbf{w}$: $\mathbf{w}^{new} = \mathbf{w}^{old} - \eta \frac{\partial}{\partial \mathbf{w}} R_{emp}(\mathbf{y}, \mathbf{x}, \mathbf{w}, b)$.

Note that $R_{emp}(\mathbf{y}, \mathbf{x}, \mathbf{w}, b) = \frac{1}{2} \sum_{n=1}^N (y^n - (\mathbf{w}^T \mathbf{x}^n + b))^2$

Select one:

- ☐ a. $b^{new} = b^{old} - \eta \frac{1}{2} \sum_{n=1}^N ((\mathbf{w}^T \mathbf{x}^n + b) - y^n) \mathbf{x}^n$
- ☐ b. $b^{new} = b^{old} - \eta \frac{1}{2} \sum_{n=1}^N ((\mathbf{w}^T \mathbf{x}^n + b) - y^n)$
- ☐ c. $b^{new} = b^{old} - \eta \sum_{n=1}^N ((\mathbf{w}^T \mathbf{x}^n + b) - y^n) \mathbf{x}^n$
- ☐ d. $b^{new} = b^{old} - \eta \sum_{n=1}^N ((\mathbf{w}^T \mathbf{x}^n + b) - y^n)$

2

Logistic regression multiple choice questions

3

Which MLP can solve the XOR problem with a sufficient number of parameters?

$f = \text{ReLU}$

- A. $W_2 f(W_1 x)$
- B. $W_2 W_1 x$
- C. $f(W_2 W_1 x)$
- D. $f(W_2 f(W_1 x))$

I think these were the different option, but I'm not a 100%

4

Write down the recursion formula for delta-errors of the backpropagation algorithm using the notation of the lecture.

$$\delta^{l-1} = \dots$$

Select one:

- ☐ a. $\delta^{l-1} = \delta^l \mathbf{W}^{l-1} \text{diag}(f'(s^{l-1})) f(s^{l-1})$
- ☐ b. $\delta^{l-1} = \delta^l \mathbf{W}^{l-1} \text{diag}(f'(s^{l-1})) s^{l-1}$
- ☐ c. $\delta^{l-1} = \delta^l \mathbf{W}^l \text{diag}(f'(s^{l-1}))$
- ☐ d. $\delta^{l-1} = \delta^l \mathbf{W}^l \text{diag}(f'(s^l))$
- ☐ e. $\delta^{l-1} = \delta^{l-1} \mathbf{W}^{l-1} \text{diag}(f'(s^{l-1}))$

5

For training a machine learning model $g(\mathbf{x}; \mathbf{w})$ with stochastic gradient descent, the important quantity to obtain updates is the gradient of the loss function $\mathcal{L}$ with respect to the inputs $\mathbf{x} : \nabla_{\mathbf{x}} \mathcal{L}(y, g(\mathbf{x}; \mathbf{w}))$, True or False?

Select one:

- ☐ a. False, the important quantity is the gradient of the weights w.r.t. the inputs
- ☐ b. False, the update is the loss scaled by the learning rate
- ☐ c. True, the gradient w.r.t. the inputs determine the direction in which to go on the loss function to minimize the loss
- ☐ d. False, the important quantity is $\nabla_{\mathbf{w}} \mathcal{L}(y, g(\mathbf{x}; \mathbf{w}))$
- ☐ e. True, the weights of the model are updated with the gradient of the loss function w.r.t. the input

6

Mark all activation functions that map all inputs to strictly **positive** values!

Select one or more:

- ☐ a. Linear
- ☐ b. SELU
- ☐ c. ReLU
- ☐ d. Sigmoid
- ☐ e. Tanh

7

Question about the derivative of certain activation functions (multiple choice), e.g.:

- $\partial \text{Sigmoid} \in [0, 1]$
- $\partial \text{Tanh} \in [0, 1]$
- $\partial \text{Sigmoid}$ is either 1 or 2
- $\partial \text{Tanh} = \tanh(1 - \tanh)$

There were more options, but I can't recall all

8

Question about the activation functions in the last layer of MLP (multiple choice) e.g.:

- **For logistic regression we often use sigmoid**
- **For multiclass classification we often use Softmax**

There were more options, but I can't recall.

9

Forward pass and backward pass calculation. The activation function was tanh (the tanh and the derivative of the tanh was listed as well)

10

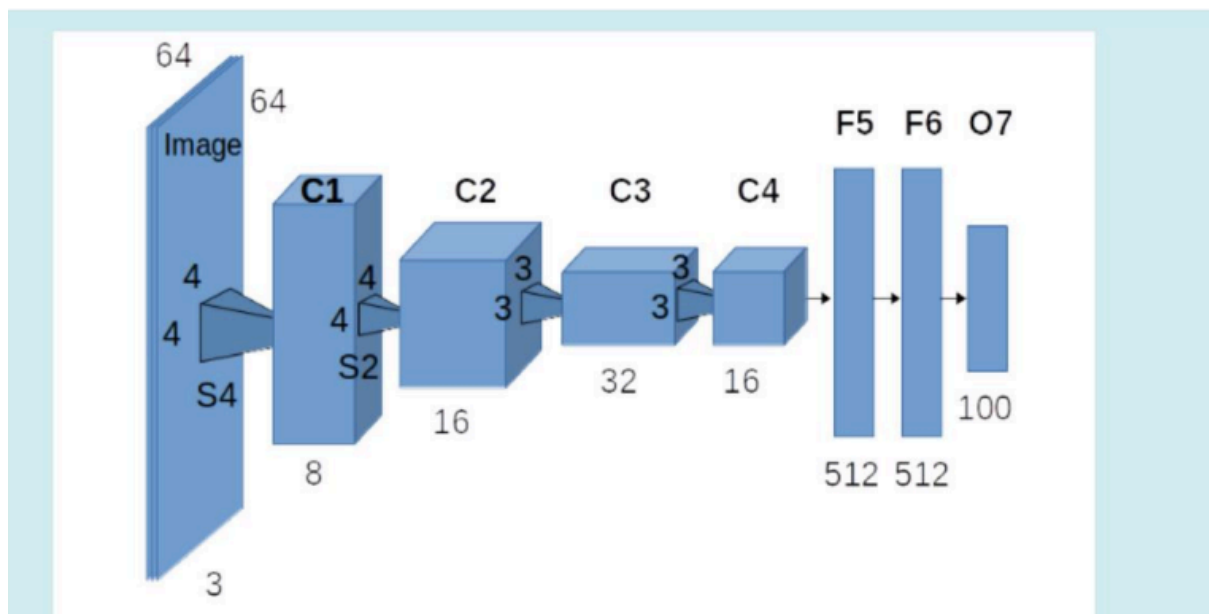
Question about cross validation (multiple choice)

11

Convolution calculation

12

Dimensions calculation



13

Calculation of all the network weights combined from the previous exercise

14

Tiles calculation (I think it was exactly the same as this one)

Image 1

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Image 2

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Pick a set of kernels (columns a, b, c, d, e, f in the image below) and neural network that **guarantee** that the neural network predicts **the same value** for **all possible locations** and **90degree-rotations** of the green tile! Note that multiple answers can be correct!

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LeCun initialization

16

Suppose we are performing gradient descent on a quadratic objective: $R(\mathbf{w}) = \frac{1}{2} \mathbf{w}^T \mathbf{A} \mathbf{w}$ with symmetric matrix $\mathbf{A}$. The dynamics of gradient descent with learning rate η can be analyzed in terms of the spectral decomposition $\mathbf{A} = \mathbf{Q} \mathbf{\Lambda} \mathbf{Q}^T$, where $\mathbf{Q}$ is an orthogonal matrix containing the eigenvalues of $\mathbf{A}$, and $\mathbf{\Lambda} = \text{diag}(\lambda_1, \dots, \lambda_D)$ is a diagonal matrix containing the eigenvalues of $\mathbf{A}$ in descending order, where $\lambda_1 = \lambda_{\max}$ is the largest eigenvalue. Then we have

$$\mathbf{w}^t = \mathbf{Q}(\mathbf{I} - \eta \mathbf{\Lambda})^t \mathbf{Q}^T \mathbf{w}^0.$$

Based on this formula, what is the value of C such that the gradient descent iterates diverge for $\eta > C$ but converge for $\eta < C$?

Select one:

- a. $C = \frac{1}{D}$
- b. $C = \frac{2}{\lambda_{\max}}$
- c. $C = \lambda_{\max}$
- d. $C = \frac{1}{\lambda_{\max}}$
- e. $C = 1$

17

Multiple choice question about ADAM

(only question I can remember is asking about the comparison between RMSprop's parameters count compared to ADAM)